

HOUSE PRICE REGRESSION

KNIME Project 2 - Team 2

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Feature Engineering

01 Handling missing values

Missing values were replaced using appropriate strategies.

This prevents model errors and improves data consistency.

02 One-Hot Encoding

Categorical variables were converted into numeric format.

This allows machine learning models to process categorical data effectively.

03 New Feature Creation

New features were created using mathematical transformations.

Example: combining or transforming existing variables to capture hidden patterns.

04 Data Transformation

Some variables were scaled or transformed to improve model performance.

This helps the model learn more effectively.

Model Evaluation

Metric: RMSE (or MAE / R^2)

	Linear Regression	Random Forest	Gradient Boosted Trees	XGBoost
RMSE	63,115	27,892	28,138	31,145
MAE	51,407	17,267	17,671	18,953
R^2	0.37	0.877	0.875	0.846
Best				

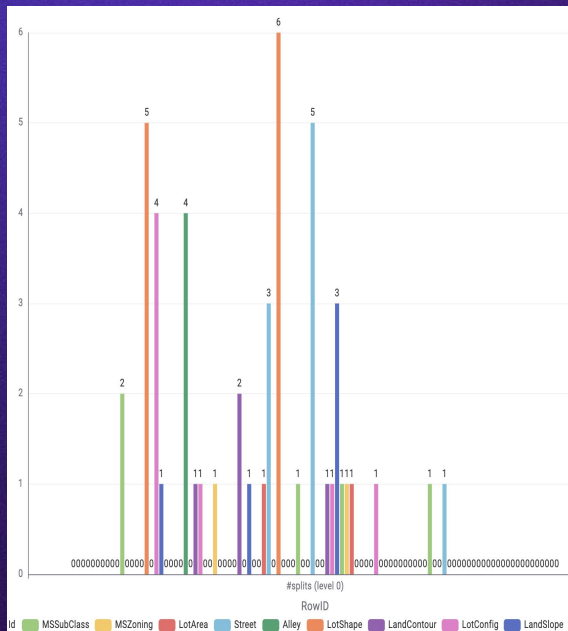
Future Improvement, Feature Importance

1. Hyperparameter Tuning

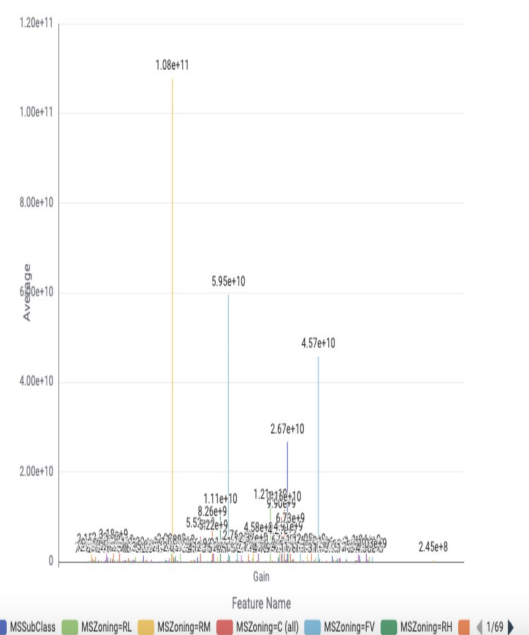
- Further tuning of model parameters could improve performance.
- For example, adjusting tree depth or learning rate.

Further improvements can be achieved by enhancing feature engineering and optimizing model parameters.

Random Forest (Split Importance)



XGBoost (Gain Importance)



Kaggle Submission & Performance

submission.csv

Id	SalePrice
1461	124439.00000000000
1462	151009
1463	186438.5685714290
1464	192251.3142857140
1465	190410.48000000000
1466	186627.33333333300
1467	171239

Leaderboard and Score

2804 Eunwoo Won



0.14925

Kaggle Submission Journey

We submitted multiple models to Kaggle to evaluate real-world performance.

Tree-based models significantly improved performance.

Hyperparameter tuning further reduced the error.